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Exploring Sleep Disorder Classification: An Evaluation of Traditional and Ensemble ML Models with XAI Interpretability

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***Abstract*— The classification of sleep disorders is important to improving human health in general. Sleep disorders and apnea may have a serious effect on human health. Manual classification of sleep stages by experts is a tiresome task, and subject to errors. In order to enhance this process, it is necessary to generate accurate machine learning algorithms to classify the sleep disorders to help in laboratory analysis, monitoring, and diagnosis of the disorders. The proposed study is a large predictive model on preliminary data and individual medical records. Machine learning algorithms like Logistic Regression (LR), LightGBM (LGBM), Gradient Boosting (GB), Random Forest (RF), AdaBoost (AB) and Decision Tree (DT) were used to execute the classification. Utilizing XGBoost (XGB) as the feature selection technique improved overall performance and enhanced feature extraction. The highest accuracy is obtained by using stacking ensemble models of RF, XGB, and LR with the accuracy being 97.33% with Precision, Recall, F1-score of 95.69%. Further, methods of Explainable Artificial Intelligence (XAI), such as SHAP and LIME, were applied to measure the importance of features and also as a way of making decisions transparent, therefore enhancing model interpretability and clinical applicability.**

Keywords*— *Sleep disorder, Stacking ensemble Model, Explainable AI, Machine Learning.

I. INTRODUCTION

Sleep is important with regard to the body and mind; it promotes brain activity, memory and good health. The inability to sleep well may alter thinking capacity and predispose an individual to health concerns like heart disease, diabetes, and obesity. Conventionally, the stages of sleep are manually measured using polysomnography (PSG) by medical practitioners which is not only time consuming but also prone to human error [1,2]. A survey of more than 13,000 adults in 13 countries by Philips in 2021 on the World Sleep Day showed that only 55 percent of adults were satisfied with the quality of sleep. Most of the problems that were reported include insomnia, snoring, shift-work sleep disorder as well as sleep apnea. It is worth noting that 37 percent reported the COVID-19 pandemic adversely affected sleep [1,2]. Medical experts determine sleep quality based on the following five-stage classification of sleep: wakefulness, N1, N2, N3, and REM. Wakingness is alertness and rapidity of the brain and N1 is the light sleep, which has slower brain waves. N2 and N3 are deeper sleep and N3 is the most difficult level to wake up. REM sleep is characterized by rapid movements of the eyes and the brain behaving as though awake. Each phase is critical to body and brain activity. Polysomnography is used to track these phases through EEG and ECG signals to record the brain and heart activity as we sleep [3,4,5]. Machine Learning (ML) and Artificial Intelligence (AI) are transforming the healthcare and medical informatics sectors. They lead in research on diagnosis, earlier detection of the disease, prognosis and outcome prediction [6,7].

Alongside the development of AI and ML algorithms, now, systems can be created to help accurately classify people at risk of developing sleep disorders [8]. In this research we aim at enhancing classification of sleep disorders i.e. None, Sleep Apnea and Insomnia through the application of machine learning on real-life clinical observations. The approach involves extensive data preprocessing, normalization, data balancing, selection of features, and stacking ensemble models. Also, Explainable AI (XAI) algorithms including LIME and SHAP are used to interpret and explain the proposed framework. The study offers five individual contributions to enhance accuracy of classification:

- This study proposes a realistic classification of sleep disorder based on 10 important clinical and genetic variables.
- To determine important and associated biomarkers, 9 machine learning methods were utilized.
- To maximize model performance, feature selection, normalization and the voting classifier are employed.
- The RF, LR, XGB ensemble models were superior to the rest, and the highest prediction accuracy of 97.33%.

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- To interpret model outputs and explain feature importance and decision-making, explainable AI (XAI) tools, SHAP and LIME, are utilized.

II. Related Work

The next section discusses some of the recently proposed methods for sleep disorder, which are categorized as machine learning-based approaches.

Ramesh et al. [9] used different ML algorithms such as XGB, LGBM, CB, RF, KNN, LR, and SVM to identify high pretest OSA using 1,479 records of the Wisconsin Sleep Cohort. Physical measurements and blood reports were used as clinical features, and Hyperparameters were optimized using Bayesian optimization and genetic algorithms. The accuracy, sensitivity, specificity and F1-score of the SVM model were 68.06, 88.76, 40.74, and 75.96 respectively.

Satapathy et al. [10] introduced a framework that used conventional ML algorithms (DT, KNN, RF) with deep learning-based sleep apnoea detectors to use a single-lead ECG, PhysioNet v1.0.0 dataset (70 records). The feature extraction and temporal pattern analysis were performed through hybrid CNN-DRNN architecture with dimensionality reduction by PCA. This method performed better than other algorithms, and they suggested hybrid deep neural networks to detect sleep apnoea based on ECG signals.

Han et al. [11] used supervised and unsupervised ML with gradient boosting, RF, and K-means to identify the severity of obstructive sleep apnoea on the basis of 4,014 patients (not made available publicly). Their procedures brought out accuracies of 88, 88 and 91, but the study had limitations in the form of single-centre data collection and missing values. The model developed gives an effective method of prediction of OSA severity.

Satapathy et al. [12] created a sleep-staging framework that fuses traditional ML (RF, KNN, SVM) with deep learning (CNN-LSTM) to predict sleep phases based on multimodal data. CNN obtained spatial information within the EEG, whereas LSTM was used to obtain temporal dynamics. The model was tested on the Sleep-EDF database, the model had a high accuracy of 87.4% compared to other algorithms. A Butterworth filter was applied to eliminate data noise.

Zhang et al. [13] designed an automated system based on a bed-integrated RF sensor to detect and predict sleep breathing problems. Near-field coherent sensing (NCS) was used to collect nighttime recordings of 27 patients in the Weill Cornell Center Sleep Medicine. The data was analyzed using a random forest model with 88.6% sensitivity and 89.0% specificity in detecting apneic events, and 81.3% sensitivity and 82.1% specificity in predicting apneic events within 90 seconds.

Warunlawan et al. [14] examined how sleep habits and lifestyle affect life quality with the help of the Sleep Health and Lifestyle Dataset. The most efficient feature selection method was MRMR, which emphasizes physical activity level, systolic blood pressure, and BMI. Bagged Trees had the highest accuracy of 91.90 out of 21 classifiers.

Table I presents a prior research with datasets and ML model. There were nine applied ML algorithms, with stacking ensemble and voting being used to determine the most efficient model and overcome restrictions. Our method works with the set of data to diagnose sleep disorders. By leveraging these algorithms, we aim to improve the accuracy and efficiency of the diagnosis process, ensuring reliable results.

TABLE I. Overview of algorithms, datasets, and accuracies

References	Models	Accuracy	Datasets
Ramesh et al. 2021 [9]	XGB, LGBM, CB, RF, KNN, LR and SVM.	SVM 68.06%.	Wisconsin Sleep Cohort dataset.
Satapathy et al. 2021 [10]	DT, KNN, RF.	89.10%, 89.10%, 94.46%.	ISRUC%-Sleep database
Han et al. 2023 [11]	gradient-boost and RF and KNN	88%, 88%, 91%	Medical Centre
Satapathy et al. 2023 [12]	CNN+LSTM, RF, KNN and SVM	87.4%, 74.07%, 83.65%, 76.04% .	Wisconsin Sleep Cohort dataset.
Zhang et al. 2023 [13]	Random Forest (RF)	Sensitivity: 88.6%	Weill Cornell Center.
Warunlawan et al. 2023[14]	Bagged Trees	91.90%	Sleep Health and Lifestyle Dataset

III. Materials And Methods

Fig. 1 is the work flow of the proposed optimized method of sleep disorder classification. This flowchart illustrates the order of procedures undertaken in the application of the classifiers and methodologies in the process of the study. Such a visual representation gives a breakdown of the steps of data processing, model training and evaluation.

A. Dataset Collection

The dataset used in this study, "Sleep Health and Lifestyle," consists of 374 samples and 13 features. This publicly available dataset can be accessed on Kaggle [15]. It includes a mix of categorical and numerical variables. Among the 13 features, five are categorical, representing distinct, non-numerical attributes that describe qualitative aspects of the data. The remaining eight features are numerical, providing continuous quantitative data. Since the 'Person ID' attribute contains unique identifiers that do not contribute to the classification task, it was excluded from the dataset during the initial preprocessing stage.The data given represents a multiclass classification problem, the goal in which is to assign each sample to one of the fixed categories. The dependent variable of this data collection is Sleep Disorder and it includes the following categories: None, Sleep Apnea, and Insomnia. Table II shows the quantity of samples of each type within the principal dataset.

B. Categorical Encoding

To measure whether there were any missing values in any of the features, we sampled the data X with the procedure in Equation (1). Having done this verification, we established that there were no missing values of any feature meaning that there were no missing values. Therefore, there was a need to perform no imputation or handling of missing values. We coded the qualitative data into numeric values to enable machine learning algorithms to act on the categorical data. To

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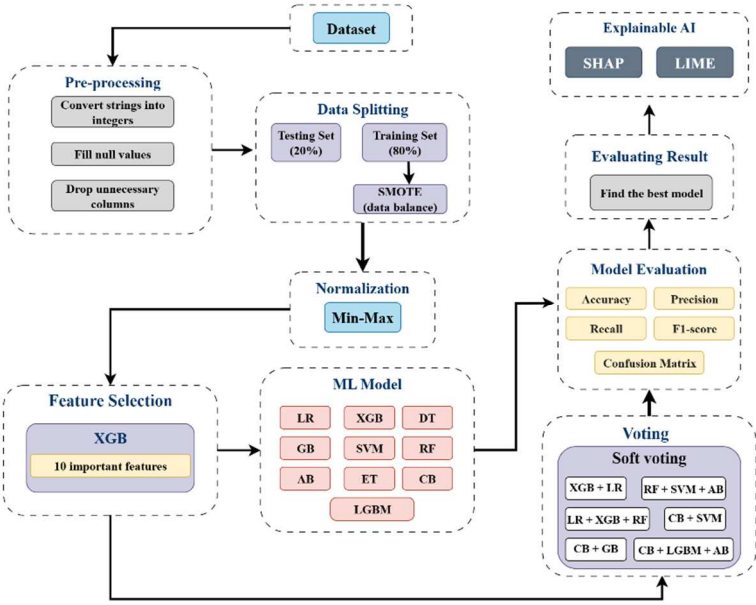


Fig. 1. Workflow of our Proposed Model

that end, we applied the label encoding [16] with the following encoded values on the sleep disorder categories: Insomnia: 0, None: 1, and Sleep Apnea: 2.

$$missing_values_count = X.isnull().sum() \quad (1)$$

TABLE II. DISTRIBUTION OF SAMPLES ACROSS CATEGORIES .

Sleep Disorder	Number of Samples
None	219
Sleep Apnea	78
Insomnia	77

C. Data Splitting

We subsequently split the dataset into training and test sets. In particular, 80% of the data was used to train. The rest 20% was left to the test set. This scheme of partitioning will enable equal representation of the data to be used in training and evaluation of the model.

D. Data standardization and normalization

In the process of feature scaling and standardization, we made use of MinMaxScaler. The values were scaled to give consistency and normalize the data as indicated in Equation (2). Here, the average of the data are brought to 0, and the standard deviation to 1.

$$N(Z) = \frac{\sum_{i=1}^{N}(Z_i - Z_{min})}{Z_{max} - Z_{min}} \quad (2)$$

where N represents the total number of data samples, Z refers to the attribute being scaled, Z_i is the value of the i-th attribute, $Z_{\min}$ is the minimum value, and $Z_{\max}$ is the maximum value of the sample.

E. Class imbalance in training sets

After train-test split, the training set samples are distributed across the categories as follows. There are 175 samples in category 1 (58.52); there are 62 samples in categories 2 and 0 (20.74). This imbalance used to be an indication of an imbalance and we have used the SMOTE technique to correct it. SMOTE produces artificial samples and eliminates the noisy samples resulting in a more balanced dataset and better performance of the model [17].

F. Feature Selection

The work of feature selection is crucial in determining the necessary qualities to reduce the dimension of the input information. In this study, we employed XGBoost in feature selection to find out the most relevant attributes by reducing the input parameters, thereby increasing the accuracy and performance of the model [18]. With reference to the feature importance scores, the first 10 features with the highest contribution to the performance of the model were chosen to be used in the proceeding analysis, to guarantee a more precise assessment of their influence on the model performance.

G. Machine Learning Classifiers

In machine learning methods, classification is extremely crucial when diagnosing diseases [19]. We have used several classifiers in this research, which are LR, DT, Extra Trees (ET), RF, XGB, AdaBoost (AB), GB, LGBM, and CatBoost (CB). These models were selected to cover both linear and non-linear decision boundaries and to provide a comprehensive comparison of widely used supervised learning algorithms for sleep disorder classification.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

The present research assesses the performance of sleep disorder classification algorithms in terms of accuracy, precision, recall, and F1 score. To improve the performance of the models, we used MinMax normalization to normalize features ranges and Synthetic Minority Over-sampling Technique (SMOTE) to balance the classes. In order to enhance predictive ability, soft voting ensemble technique

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was used that combines the results of various classifiers to make solid decisions.

A. Confusion matrix

The confusion matrix was developed so as to give comprehensive performance analysis of both binary and multiclass classification problems. To make successful predictions and classification of sleep disorders, the machine learning classifier was developed and tested on the ready dataset.

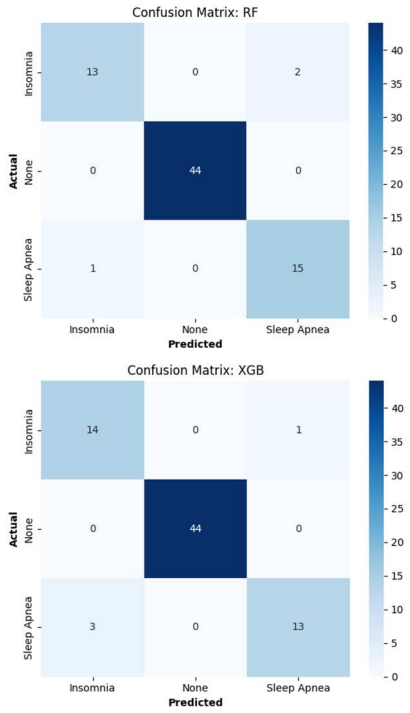


Fig. 2. ML model (top) confusion matrix.

The accuracy, precision, recall and F1-score based on the classification report and confusion matrix were used to evaluate the model. It is important to focus on recall since false negatives can result in undiagnosed and untreated disorders, which are more dangerous than false positives. The confusion matrix shows the distribution of correct and incorrect classifications of the predictions, which is given in Fig. 2.

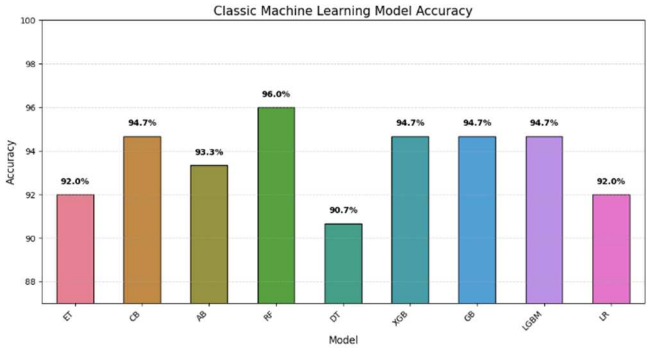


Fig. 3. Evaluation of Classic ML Model Accuracy

Multiple classical ML model is compared in order to assess the classification effectiveness of the models for sleep disorder classification. Fig. 3 shows the accuracy of various models where RF had the best accuracy with 96% followed by CB, XGB, LGBM and GB 94.7%. The accuracy values achieved were 93.3%, 92.0% and 92.0% for AB, ET and LR, respectively. Note that DT recorded the worst accuracy 90.7% and weaker generalization compared to ensemble methods.

TABLE III. EVALUATING THE PERFORMANCE OF ML MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
RF	96.00	93.70	93.47	93.52
CB	94.67	91.74	91.53	91.39
XGB	94.67	91.74	91.53	91.39
LGBM	94.67	91.74	91.53	91.39
GB	94.67	91.74	91.53	91.39
AB	93.33	89.36	89.17	89.20
LR	92.00	88.60	88.41	88.44
ET	92.00	88.85	90.01	88.90
DT	90.67	87.62	89.26	87.71

Table III also describes the performance for accuracy, precision, recall, and F1-score. RF performed best with an F1-score of 93.52%, and CB, XGB, LGBM and GB all remained over 91%. Ensemble approaches showed better performance than the simple models, including LR, ET, and DT.across all metrics, validating the robustness of the ensemble methods for the classification of sleep disorders

B. Stacking Ensemble

The efficiency of the different ensembles of classifiers for the classification of sleep disorders are shown in and summarized in Table IV The best overall performance results were obtained by XGB + LR and RF + XGB + LR ensembles with soft voting, both reaching an accuracy 97.33%, precision 95.69%, recall 95.69% and F1-score 95.69%. The overall results demonstrate the robustness of ensemble learning, with the RF + AB ensemble and the AB ensemble achieving highly similar results on all of the evaluation metrics: the RF + AB ensemble achieved an overall accuracy of 96.00%, precision of 93.70%, recall of 93.47%, and F1-score of 93.52%. Even more, CB, LGBM and GB (CB + AB, CB + ET, LGBM + GB, and LGBM + ET) all had stable performances with an accuracy of 94.67%, precision of 91.74%, recall of 91.53%, and F1-score of 91.39%. However, the GB + DT combination showed relatively poor performance with accuracy, precision, recall, and F1-score of 93.33%, 90.21%, 90.77%, and 90.12% respectively, indicating that the two classifiers are not very complementary. In all, the results highlight the superiority of ensembles. The improved performance of the RF + XGB + LR stacking ensemble is likely due to the diversity of the base learners. Random Forest captures non-linear interactions through bagged decision trees, XGBoost focuses on difficult samples via gradient boosting, and Logistic Regression provides a simple linear decision boundary. Combining these complementary strengths allows the meta-model to reduce both variance and bias, leading to higher accuracy and F1-score than any individual classifier.

Receiver Operating Characteristic (ROC) curve is one of the most commonly used graphical methods to assess diagnostic performance of classifiers. The True Positive Rate (sensitivity) is plotted against the False Positive Rate (1-specificity), and the Area Under the Curve (AUC) is used as a summary measure of model quality. An AUC near 1.0 indicates good discrimination ability while values near 0.5

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indicate random performance. Fig. 4 shows the ROC on the training set.

TABLE IV. EVALUATING THE PERFORMANCE OF ENSEMBLE MODELS

Model with Voting (soft)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
XGB + LR	97.33	95.69	95.69	95.69
RF + XGB + LR	97.33	95.69	95.69	95.69
RF + AB	96.00	93.70	93.47	93.52
CB + AB	94.67	91.74	91.53	91.39
CB + ET	94.67	91.74	91.53	91.39
LGBM + GB	94.67	91.74	91.53	91.39
LGBM + ET	94.67	91.74	91.53	91.39
GB + DT	93.33	90.21	90.77	90.12

DT, ET, RF, XGB, GB, LGBM and CB all had high AUC values (i.e. high separation capability) of 0.98. The resultant AUC for AB was 0.96 and for LR it was a little lower at 0.94. These results show that ensemble-based models generally do better on training data than linear models. Fig. 5 shows the ROC curves on the test set. With an AUC of 1.00 RF gave the best result, meaning almost perfect classification. Other models like LR, XGB, GB, LGBM and CB gave similar results with AUC 0.98. The generalization ability of ET reached an AUC of 0.97, while the generalization ability of DT was subpar with an AUC of 0.93. These results validate that ensemble methods retain good predictive performance on test data while simpler models such as DT tend to lose performance.

C. *Explainable AI (XAI)*

We used our best two model agnostic explainability methods: SHAP and LIME to provide transparency to the sleep disorder classification. These approaches offer both global and local interpretability, thereby allowing the understanding of the contributions made by physiological and behavioral features to the predictions. SHAP and LIME were applied to the final stacking ensemble (RF + XGB + LR), treating it as a black-box model. All reported explanations and feature importance results correspond to this ensemble. Fig. 6 shows the SHAP beeswarm plot, that indicates the contribution of each feature to the model output globally. Of all predictors, physical activity level, blood pressure and age were the most significant. The SHAP values show the direction and magnitude of the effect of these features on classification, whereas the color scale shows the feature values. This global view is necessary for the identification of the most relevant health indicators in the dataset. Such patterns of importance have clinical meaning. Low physical activity is linked to poor quality of sleep and high risk of insomnia and sleep apnea as being sedentary can exacerbate weight gain, cardiovascular strain and stress. Elevated blood pressure is strongly associated with obstructive sleep apnea, in which repeating interruptions to breathing and oxygen desaturations have been implicated in the development of hypertension and cardiovascular risk. Age is also a known risk factor as the prevalence of insomnia and sleep apnea increases in middle-aged and older adults. Increased stress levels, reduced sleep time and increased BMI categories strengthen these associations even further, given that chronic stress, sleep restriction and obesity are all well-documented contributors to sleep fragmentation and respiratory disturbances while sleeping. Therefore, the SHAP results are

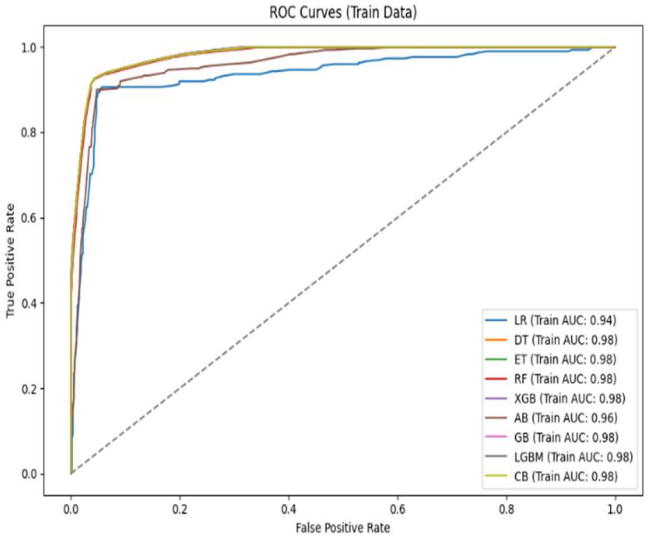


Fig. 4. ROC curve and AUC of train data

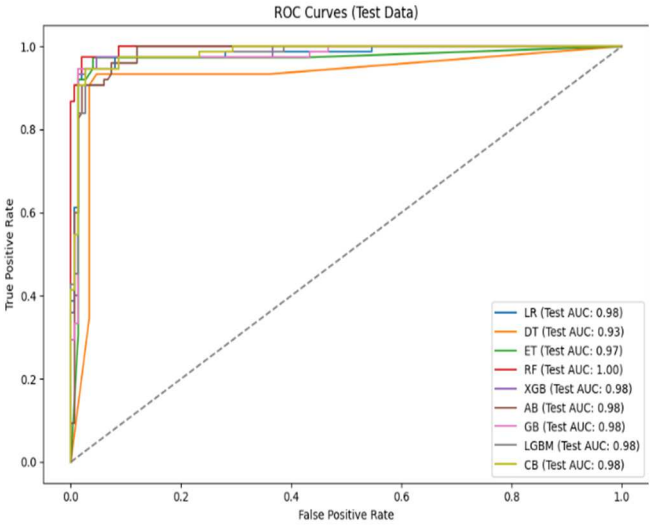


Fig.5. ROC curve and AUC of test data

in good agreement with the known clinical knowledge rather than spurious correlations. Fig. 7 is a LIME explanation for one prediction instance. The chart shows the probability distribution over three outcome classes (Negative, Neutral, Positive) and the most important contributing features. LIME thus offers local interpretability through an approximation of the model with simpler surrogate functions.

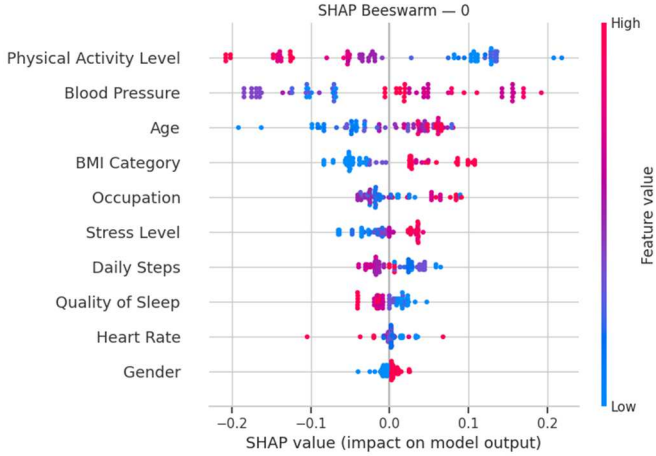


Fig. 6. Explainability using SHAP

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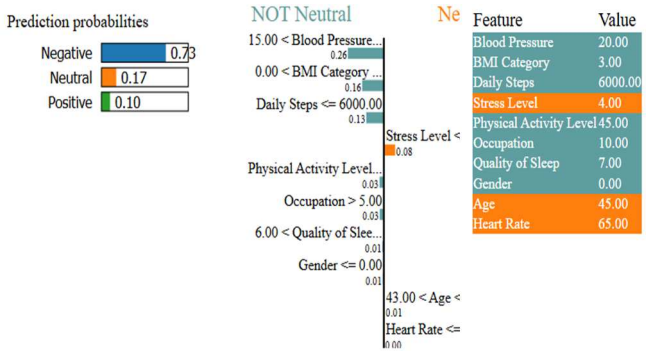


Fig. 7. Explainability Using LIME

Finally, Figure 8 also visualizes feature values of the same subject used in the LIME interpretation. This table links raw inputs (age = 45 years, heart rate = 65 bpm, stress level = 4) to interpretability, so that clinicians can track decisions all the way back to the parameters of the patient. The LIME explanations for individual subjects are consistent with the global SHAP analysis: subjects with high stress, lower physical activity, high blood pressure and shorter sleep duration are more likely to be classified as having a sleep disorder, which is consistent with what is known from current clinical understanding. Together, SHAP and LIME increase model transparency, build clinical trust and improve usability of AI-based diagnostic systems.

Table V shows that our method shows better performance in sleep disorder classification than the older studies. Ramesh et al. [9] with SVM had only 68.06% accuracy, whereas the performance of Han and Oh [11] with LightGBM was 91.11% but had lower recall (75.66%). Warunlawan et al. [14] used MRMR with Bagged Trees and showed balanced measures of about 91.9. By contrast, our ensemble model (RF + XGB + LR) had an accuracy of 97.33% with a precision of 95.69, a recall of 97.33, and F1-score of 95.69. Also, our framework is highly accurate and clinically transparent due to the integration of SHAP and LIME.

TABLE V. COMPARING PREVIOUS WORK WITH OURS

Ref.	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	XAI
[9]	SVM	68.06	66.36	88.76	75.96	No
[11]	LGBM	91.11	90.87	75.66	70.19	No
[14]	Bagged Trees	91.90	91.95	91.90	91.93	No
Our Work	RF + XGB + LR	97.33	95.69	95.69	95.69	SHAP, LIME

V. CONCLUSION

This work presents a machine learning framework which improves the accuracy and reliability of sleep disorder classification. By applying ensemble strategies such as Random Forest, XGBoost, and Logistic Regression with soft voting, the model achieved strong performance, recording an accuracy of 97.33% with precision, recall, and F1-score values of 0.9569. Moreover, XGBoost feature selection, MinMax normalization, and SMOTE were used for feature extraction, feature scaling, and class imbalance, respectively. Besides the predictive power, we ensured explainability with SHAP and LIME, which gave explicit insights into the most influential physiological and behavioral drivers. These explainable findings are more trustworthy in clinical practice, and allow matching of model predictions to medical domain expertise. Overall, the proposed framework shows that ensemble learning can be coupled with explainable AI to provide a robust, scalable, and transparent system for accurate diagnosis and monitoring of sleep disorders.

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